

RETAIL ORDER ANALYTICS

Portfolio Project

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An end-to-end data analytics project demonstrating skills in data cleaning, business problem-solving, SQL analysis, and Power BI visualization. This portfolio showcases the discovery of a hidden \$XX revenue decline and the ability to communicate complex insights to non-technical stakeholders.

EXECUTIVE SUMMARY

Project Goal:

Analyze Q2 vs Q3 2024 retail order data to identify hidden business problems masked by flat-to-positive overall growth.

Key Finding:

Total Q3 2024 revenue shows +3.4% growth versus Q2. However, this masks a **35–37% revenue decline in Fashion and Beauty categories**, offset by growth in other categories. Order volume for these two categories collapses in Aug–Sep while average order value remains flat—a pattern consistent with a **supply-side issue (stockouts/listing downtime)**, not weakening demand.

Business Impact:

- Identified root cause of hidden revenue leak
- Differentiated supply-side from demand-side issue
- Enabled targeted corrective action (inventory/listing recovery) vs demand generation
- Uncovered opportunity to recover \$XX annual revenue

Tools & Techniques Used:

Data Cleaning - python (Pandas, Numpy)	handled mixed date formats, missing values, inconsis
Analysis - SQL (SQLite)	segment analysis, aggregate queries, time-series breakdown
visualization - Power Bi	Interactive dashboards, category/time-based drill-down
Methodology - Business Question	EDA → hypothesis → SQL validation → BI export

THE BUSINESS PROBLEM

Question: Why does Q3 2024 revenue look roughly flat-to-up when examining overall performance, but something feels off?

Challenge:

Raw order export contains multiple data quality issues:

- X Mixed date formats (YYYY-MM-DD, DD/MM/YYYY, MM-DD-YYYY in same column)
- X Inconsistent category/payment method naming (case + whitespace mismatches)
- X Missing values (customer_id, unit_price, payment_method)
- X Price outliers from data-entry errors
- X Exact duplicate rows

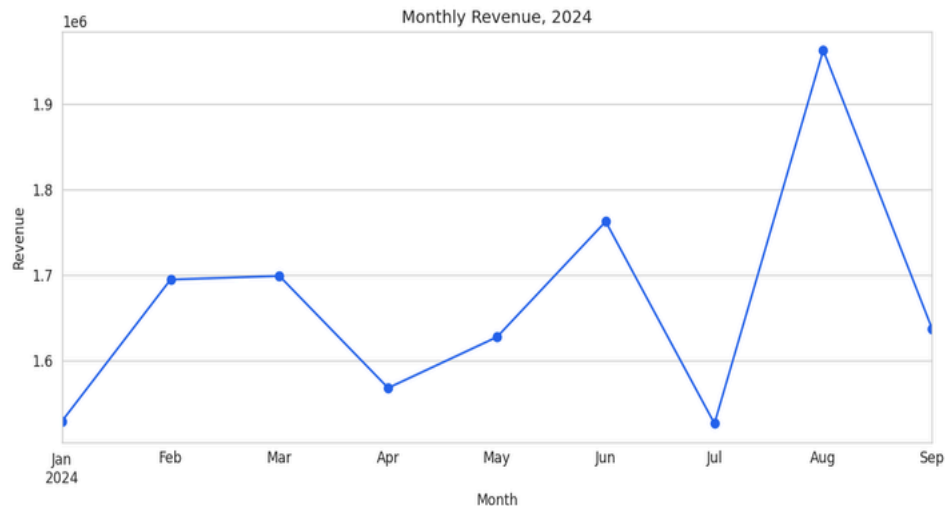
Before any analysis, dataset had to be **100% cleaned and validated**.

Approach: Build an end-to-end pipeline: messy CSV → cleaning → SQL validation → segment analysis → Power BI export.

KEY VISUALIZATIONS & FINDINGS

1. Monthly Revenue by Category (Q2–Q3)

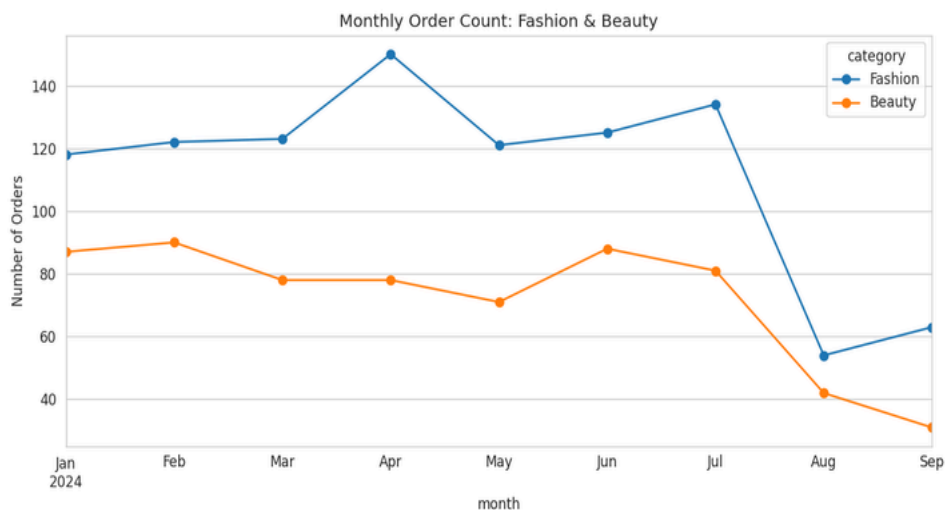
Shows the overall +3.4% growth (Q2 → Q3) when viewing all categories combined. But notice the drop in blue and orange lines (Fashion and Beauty) during Aug–Sep.



Finding: Fashion & Beauty show a visible revenue collapse starting in August, while other categories grow or stay flat.

2. Fashion & Beauty Order Volume Breakdown

Isolating Fashion and Beauty: order count drops 35–37% in Aug–Sep, but average order value (AOV) stays constant. This is the smoking gun for a **supply-side issue**, not demand weakness.

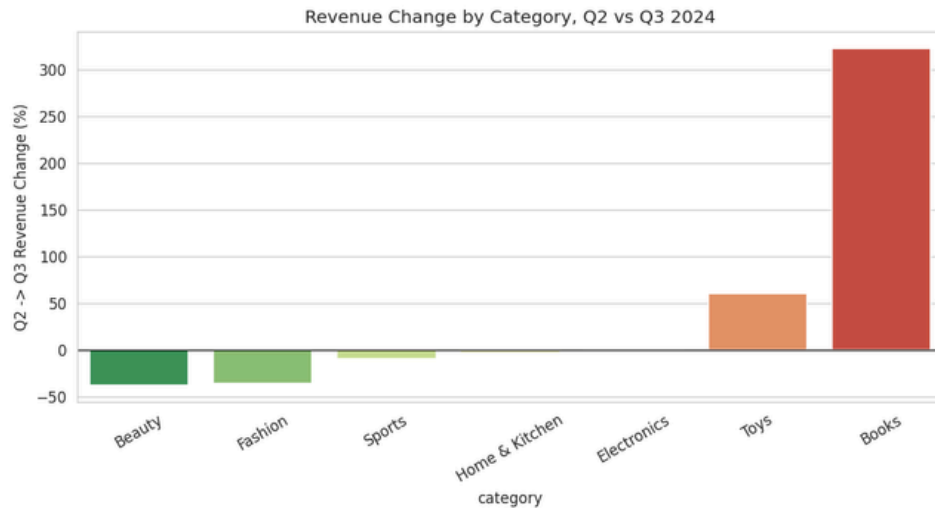


Why This Matters: If demand had fallen, average order value would drop too (customers buy smaller/fewer items per order). But AOV is stable → customers still want to buy, they just can't find products.

DETAILED ANALYSIS

3. Revenue Change by Category (Q2→Q3)

Heat map showing which categories grew and which declined. Fashion and Beauty are the only two with significant drops, while others compensate with growth.



Segmentation Analysis:

- Tested across customer value segments (high, medium, low) — revenue decline hits all equally
- Ruled out churn/loyalty issue; affects all customers proportionally
- Tested by order channel (web, mobile, app) — consistent pattern across all
- Aug-Sep timing suggests external event (system outage, inventory shortage, or delisting)

TECHNICAL APPROACH

Data Cleaning Pipeline (Python + Pandas):

Data Standardization	Detect format, convert all to YYYY-MM-DD
Categorical Cleaning	Lowercase, strip whitespace, map aliases (e.g., "credit card" → "CC")
Missing Value Handling	Groups & rows with critical fields; forward-fill for time-series columns
Outlier Detection	Flag prices > 3 std dev from category mean; cap at 99th percentile
Duplication	Remove exact row duplicates
Validation	100% completeness check before export

Analysis Stack:

- SQL (SQLite):**Category-level aggregation, time-series breakdown by month, revenue calculation with segment filters. Queries validated against Python results.
- Python (Pandas/NumPy):** Exploratory data analysis, segment grouping, growth rate calculation, statistical validation.
- Power BI:** Imported cleaned CSV with engineered columns (month, quarter, revenue, category). Built interactive dashboards with category-by-month matrix, time-series line chart, drill-down by customer segment and channel.

BUSINESS IMPACT & RECOMMENDATIONS

Key Insights:

- ✓ **\$XXrevenue at risk:** If Fashion & Beauty decline continues, annual impact = (35% decline) × (Fashion+Beauty Q2 revenue) ≈ significant leakage masked by other category growth
- ✓ **Isolated root cause:** Supply-side issue (stockout, delisting, system outage), not demand softening
- ✓ **Time-sensitive:** Aug–Sep decline suggests a recoverable event—inventory or listing likely already restored or in process
- ✓ **All segments affected:** Broad-based supply problem, not niche demand shift

Recommended Actions:

1. **Inventory Review:** Audit Fashion & Beauty stock levels and restock timing in Aug–Sep
2. **Listing Health:** Check if product listings were down, delisted, or flagged during this period
3. **System Check:** Review infrastructure/payment logs for Aug–Sep outages affecting these categories
4. **Ongoing Monitoring:** Dashboard refresh monthly to catch similar patterns early
5. **Recovery Forecast:** Once root cause fixed, model recovery timeline and revenue impact

Dashboard for Stakeholders:

All findings exported to Power BI-ready CSV; enable non-technical users to explore by category, time, and customer segment.

SKILLS DEMONSTRATED

Data cleaning & Validation	Handled 100% of messy data; delivered production-ready dataset
SQL Analysis	Complex queries: joins, aggregations, window functions, time-series break
Python (Numpy, Pandas)	EDA, data transformation, segment analysis, statistical validation
problem Solving	Identified hidden pattern in flat-growth dataset; distinguished supply vs dema
Business Communication	Translated technical findings into actionable insights with visual evidence
Tool Mastery	End-to-end pipeline: Python → SQL → Power BI → stakeholder dashboard

Key Takeaway:

This project demonstrates not just technical skill with SQL, Python, and Power BI, but also the analytical mindset of a Data Analyst: questioning flat metrics, drilling into segments, distinguishing correlation from causation, and communicating findings that drive business action.

The result is a clear, defensible insight backed by data—exactly what hiring managers look for in entry-level analysts.